Masting Project – UBC Visit

January–March 2026 — Analysis Plan (Draft)

People involved: Victor, Janneke, Lizzie, and Éléonore *(Am I forgetting someone? Is Mao involved?)*

Study system: Mount Rainier National Park

Dataset: Seed-fall data (multi-species, multi-site, multi-year)

Research Questions and Hypotheses

1) Do species exhibit interspecific synchrony in seed production at Mount Rainier National Park?

Hypothesis: Preliminary results suggest interspecific synchrony among several species.

Approach:

- For each site, calculate yearly mean seed production per species.
- Quantify synchrony using the variance ratio (Loreau & de Mazancourt) applied to species-level time series.
- Compare observed synchrony against null expectations?

2) Is seed production synchronized within species (intraspecific synchrony) across sites at Mount Rainier?

Approach:

- For each species, compare annual seed production across sites.
- Apply the variance ratio to multisite time series for that single species.
- Compare synchrony levels between species to see which species show tighter intraspecific coupling.

3) Is total resource availability (g/m^2) temporally synchronized from the perspective of seed predators?

Note: Synchrony in total resource availability may drive predator responses and link to the seed predator satiation hypothesis.

Approach:

- Define a metric for resource availability (e.g., annual total seed fall per m^2 or a nutritional index if species differ strongly in energetic value).

- Identify years of low vs. high total availability (resource pulses).
- Explore how these pulses might relate conceptually to chapters 2 and 3 (predation and predator satiation), without going too far analytically yet.

4) (Potential) How does interspecific synchrony vary along the elevational gradient?

Approach:

- Estimate interspecific variance ratios separately for each elevation band (site).
- Examine whether synchrony strengthens or weakens with elevation.

5) (Potential) How does intraspecific synchrony vary with elevation, and can these patterns be linked to seed-predator home-range sizes?

Approach:

- Quantify intraspecific synchrony for each species across elevations.
- If differences emerge, explore whether predator movement ecology (e.g., home-range size of rodents, corvids, squirrels) could interact with the scale of resource pulses.

Key Points to Clarify

- **Terminology:** We are not defining masting here; instead, we treat species as masting or quasi-masting based on known ecology. We can either:
  - use "synchrony in seed production" throughout, **OR**
  - refer to "masting synchrony" without formally quantifying masting itself (unless CV is needed).
- **Modeling:** Adapt Victor's model to handle multiple species jointly for interspecific analyses.
- **Resource availability metric:** Must be defined early—simple biomass (g/m^2) may be sufficient, but consider adjusting for seed energy content if relevant.